Changing velocity

antocciphysics | TEACHER KEY | CPA

Learning target: Calculate acceleration and distinguish speeding up, slowing down, and turning.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 4; 3 course days later.*
Displacement -24 m. Second arrow points southeast with component ratio 8: 3.

02. Explain a worked example (worked)
Both velocity arrows east, final one shorter; acceleration west.

03. Watch velocity grow (model)
8-2=6 m/s; 6/3=2 m/s²; speeding up.

04. A bike slows (practice)
Loses 6/5=1.2 m/s each second; acceleration 1.2 m/s² west, opposite motion.

05. A turn at steady speed (reason)
Yes. Tangent velocity directions change, so velocity changes.

06. Choose a method (mixed) *Review: Unit 1, Day 6; 1 course days later.*
3 m east and 4 m south. Add each pair of signed components.

07. Exit check (exit)
Slowing down; velocity changes 4 m/s east over 2 s, so 2 m/s² east.

08. STRETCH: Apply it another way (stretch)
Graph from -1 to -5 m/s; slope -2 m/s². Speed increases although acceleration is negative.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 4; 3 course days later.*
Displacement 12 m; distance 36 m. An angle needs a reference direction/axis and rotation sense.

02. Explain a worked example (worked)
Both velocity arrows east, final one shorter; acceleration west.

03. Signed acceleration (model)
a=(4-(-8))/3=+4 m/s²; slows untilt=2 s, rests instantaneously, then speeds up positive.

04. Braking intervals (practice)
(a) a = (0 - 24)/4.0 = -6.0 m/s². (b) Average velocity = (24 + 0)/2 = 12 m/s; displacement = 12 × 4.0 = +48 m. (c) Acceleration is opposite the positive motion, so it is negative. The car keeps moving forward while braking, so displacement is positive.

05. Small acceleration, long time (reason)
Δv=2 m/s. Accumulated change depends on time; this is a hypothetical comparison, not a real asteroid forecast.

06. Choose a method (mixed) *Review: Unit 1, Day 6; 1 course days later.*
R=(2, 4) m; magnitude ≈4.472 m at 63.43° north of east. C=(-2, -4) m.

07. Exit check (exit)
First -3 m/s², speeding up. Then 0 m/s², constant speed 9 m/s.

08. STRETCH: Apply it another way (stretch)
Turns at 3 s. Displacement -30+25=-5 m. Distance 9 m west plus 4 m east=13 m.